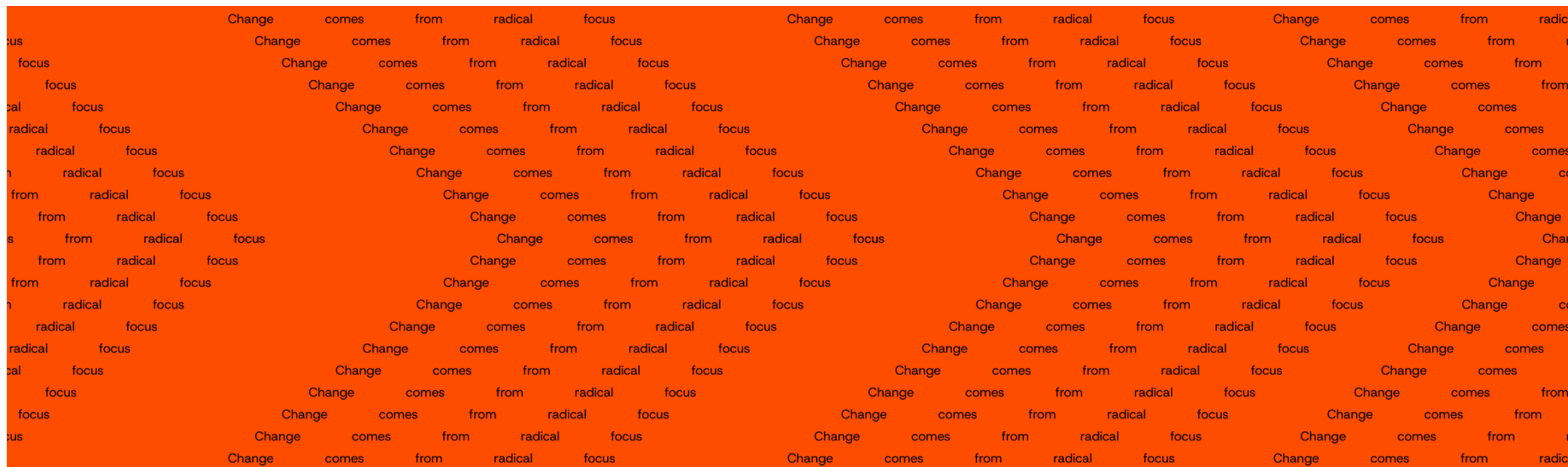




ThingWorx Proficiency Historian API Building Block

Eliot Landrum
September 2026

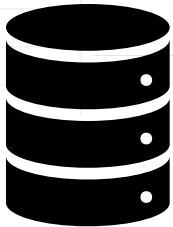
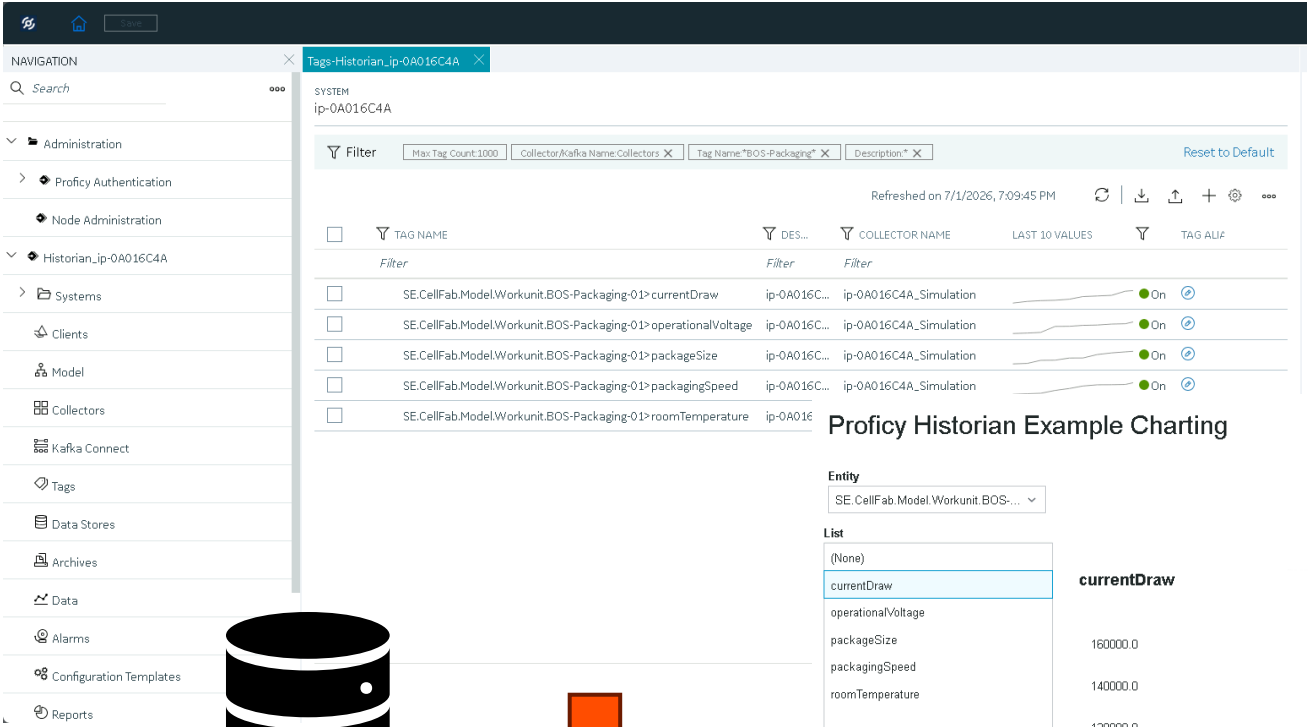


Purpose

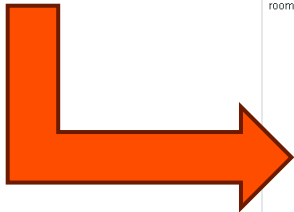
Demonstrate Proficy Historian API connectivity to ThingWorx

- >Proficy Historian has a [native REST API interface](#) for querying time series data and information about the available tags.
- >This Building Block is a proof of concept demonstration to authenticate and query time series data from Proficy Historian

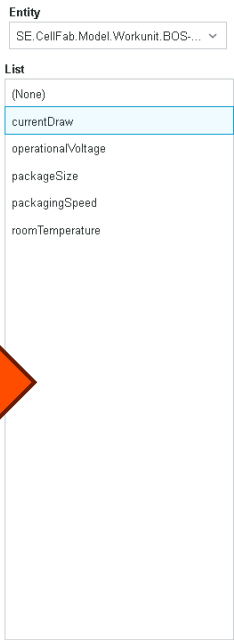
Front-End Results



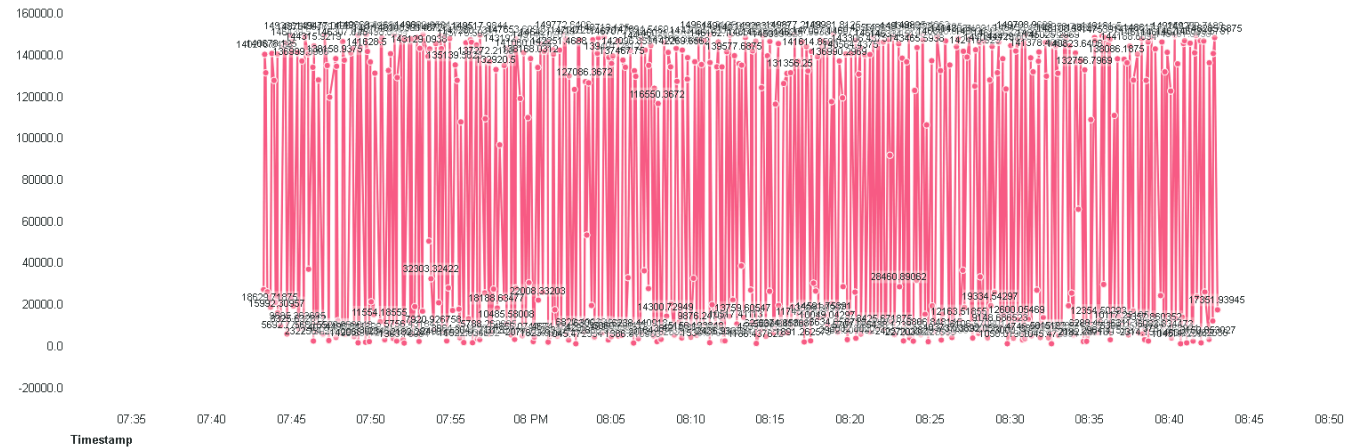
Proficy Historian
Configured with
simulation data



Proficy Historian Example Charting



currentDraw



ThingWorx
Example Chart Mashup with Proficy
Historian data

Configuration

Thing:TWSC.ProficyHistorianAPI.Manager ⓘ

To Do ▼

Save

Cancel

More ▼

General Information ⓘ

Properties and Alerts ⓘ

Services ⓘ

Events ⓘ

Subscriptions ⓘ

Configuration ⓘ

Permissions ⓘ

Change History ⓘ

View Relationships ⓘ

Configuration

All Configurations

+ Add

Refresh

The Building Block manager holds the Request Configuration end points. Any Proficy Historian API calls can be added using the template format that Swagger uses.

RequestConfiguration

RequestConfiguration (5) + Add

Actions	RequestDisplayName	RequestType	RequestSpecification	Category
 	Get tags	GET	/v1/tags/	Tags
 	Get tags with name mask	GET	/v1/tags/{nameMask}/{maxNumber}	Tags
 	Get raw data	GET	/v1/datapoints/raw?tagNames={tagNames}&start={start}&end={er	Data
 	Get interpolated data	GET	/v1/datapoints/interpolated?tagNames={tagNames}&start={start}&e	Data
 	Get tag properties	GET	/v1/tags/properties/{tagName}	Tags

ThingConfiguration

ThingConfiguration + Add

Actions	id	thingName
No data		

ServerConfiguration

ServerURL

https://historian-server:8443/historian-rest-api

TokenURL

https://historian-server:8443/uaa/oauth/token

Username

historian_api_user

Password

***** 

Connection information is also configured in the Building Block Manager and requires the Server URL, Token URL, and a REST API-enabled user credentials (configured in Proficy administration).

Administration – Execute Requests

After an API Request is configured, the “Execute Requests” page can be used to execute the request and view the raw JSON response from the API. Input Parameter text fields are dynamically shown.

If any POST data needs to be provided, a collapsible input box is on the far right.

Execute Proficy Historian API Requests

API Requests

Get raw data

tagNames

SE.CellFab.Model.Workunit.BOS-Packag

start

2026-09-23T19:43:06.837Z

end

2026-09-23T20:43:00.000Z

count

500

direction

1

Execute Request

API Results

```
{
  "Data": [
    {
      "TagName": "SE.CellFab.Model.Workunit.BOS-Packaging-01>currentDraw",
      "Samples": [
        {
          "DataAttributes": [],
          "Quality": 3,
          "Value": "41115.91016",
          "TimeStamp": "2026-09-23T19:43:07.000Z"
        },
        {
          "DataAttributes": [],
          "Quality": 3,
          "Value": "30029.69531",
          "TimeStamp": "2026-09-23T19:43:08.000Z"
        },
        {
          "DataAttributes": [],
          "Quality": 3,
          "Value": "70636.71094",
          "TimeStamp": "2026-09-23T19:43:09.000Z"
        },
        {
          "DataAttributes": [],
          "Quality": 3,
          "Value": "86415.69531",
          "TimeStamp": "2026-09-23T19:43:10.000Z"
        }
      ]
    }
  ]
}
```

> API Request Body (if required)

Manager Thing Shape Services

Using the Configuration Tables as a reference for the requests, the services can flexibly call any Proficy Historian API and return JSON data.

The Manager automatically handles OAuth token fetching, caching, and renewal.

Thing Shape:TWSC.ProficyHistorianAPI.Management_TS

To Do

Save

Cancel

More

General Information

Properties and Alerts

Services

Events

Subscriptions

Configuration

Permissions

Change History

View Relationships

Services

Filter by name

Choose entity

Choose category

All Services

Add

Duplicate

Browse Remote Services

Delete

(showing 11 of 11)

	Entity	Name	Actions	Execute	Access Modifier	Category	Type	Inputs	Output
☐	Me	ExecuteRequest	 			API	Local (JavaScript)	-T: Parameter1 -T: Parameter2 -T: Parameter3 -T: Parameter4 -T: Parameter5 -T: Parameter6	 result
☐	Me	GetHistorianEntities	 			Historian Tags	Local (JavaScript)		 result
☐	Me	GetHistorianEntityTags	 			Historian Tags	Local (JavaScript)	-T: entityName	 result
☐	Me	GetHistorianTagMax	 			Historian Tags	Local (JavaScript)	-T: thingName -T: propertyName	# result
☐	Me	GetHistorianTagMin	 			Historian Tags	Local (JavaScript)	-T: thingName -T: propertyName	# result
☐	Me	GetOAuthToken	 			API	Local (JavaScript)		-T: result
☐	Me	GetParametersFromRequest	 			API	Local (JavaScript)	-T: RequestSpecific...	 result
☐	Me	GetRequestSpecification	 			API	Local (JavaScript)	-T: RequestDisplay...	 result
☐	Me	GetRequestSpecificationWithParameters	 			API	Local (JavaScript)	-T: ParameterVal... -T: ParameterVal... -T: ParameterVal...	-T: result

Final Notes

This implementation was developed as a proof of concept and assumes:

- > This Building Block assumed that Proficy Historian tag names will be in a scheme that fits with existing assets in a demo data set. No complex manipulation or mapping table of tag names was done.
- > For the provided example services, the Historian tags must follow the `>><PropertyName>` naming convention.
- > The initial authentication token fetch can take several seconds.
 - > If this were a production system, a scheduler might be created to refresh the token behind the scenes on a periodic basis.
- > No more than six parameters are needed for request specifications
- > String tag types were not tested due to limitations of our test environment.
- > Scale testing was not performed.

Additional Historian deployments may require modifications to request definitions, authentication settings, or tag discovery logic.

